Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Python and TypeScript engineer with 7+ years shipping production systems, now building agentic AI applications with Claude Code and LangGraph daily. Understands where agents help and where they hide problems -- and builds the guardrails that enforce that line in code. Built **Leasel** -- an 8-agent HITL platform where agents reason but deterministic logic makes every regulated decision, with a CI eval harness (93.6% precision, 1,240 test cases) -- and **AgentProbe**, an open-source LLM toolkit routing across 5 providers. Model Context Protocol (MCP) in daily stack. Currently deploying Claude into production financial workflows at Intuit. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Case studies: AgentProbe · Leasel

CORE COMPETENCIES

MCP Server Development | Agent Trust Boundaries | LLM Agents (LangGraph) | Python (9+ yrs) | TypeScript / Node.js | FastAPI | RAG Pipelines | LLM Evaluation / Evals | AWS / Azure / GCP | Docker / Kubernetes | Production Guardrails | Claude Code (daily)

EXPERIENCE

Intuit

Jun 2026 - Present

Software Engineer · Virginia, USA

- Build guardrails that enforce trust boundaries around Anthropic Claude -- input validation, sensitive-data handling, and output checks so agents cannot produce unchecked financial outputs for users.
- Deploy Claude into production financial data workflows: LLM APIs, structured-output schemas, and retrieval pipelines that ground every answer in source records.
- Ship eval harnesses (LLM-as-judge + golden datasets) to block hallucinations and model quality regressions before release; CI-integrated so every PR is tested.
- Build ML pipelines with scikit-learn and XGBoost and FastAPI backends with React/TypeScript front ends; containerize with Docker and ship via GitHub Actions CI/CD to AWS.
- Use Claude Code as primary development environment daily -- understand where it accelerates work and where generated code needs careful review.

Entrata

Oct 2020 - Jul 2024

Senior Software Engineer · Multi-tenant SaaS, 100K+ monthly active users

- Owned 20+ features end-to-end (Python, PHP, Java, JavaScript, React, PostgreSQL) across 120+ Agile sprints with full production responsibility.
- Maintained the annual HUD income-limit import for 7 years -- transforming federal source data into ~180,000 domain records powering compliance workflows for 100K+ housing units; zero violations across all cycles.
- Built a RabbitMQ async processing system for rent updates across thousands of tenant records; retry logic and idempotency eliminated double-processing in production.
- Ran Python/Bash automation cutting manual ops work by 80%; built internal debugging tools reducing triage time by 40%.
- Led Docker microservices redesign (50% better scalability, 30% faster deployments); owned AWS S3 PHI migration with Libsodium encryption -- 100% compliance, zero security incidents.

Entrata India

Sep 2018 - Oct 2020

Junior Software Engineer · Pune, India

- Built affordable housing and property management modules (Python, PHP, Java, PostgreSQL); delivered large-scale data import and reconciliation workflows (1.6M HUD/Tax Credit records).
- Led AWS S3 cloud migration from legacy server storage; created 56 state-specific housing certification documents.

PROJECTS

Leasel -- Multi-Agent HITL Platform with Trust Boundaries

github.com/venkatalakshmiparimala/Lease_complaince_rag

93.6% precision, 87.9% F1 on 1,240 eval cases. 8-agent LangGraph StateGraph (triage, retrieval, reasoning, validation + HITL escalation) for lease compliance. Core design rule enforced in code: the LLM reasons, deterministic logic decides every eligibility/financial/compliance call -- agents cannot override it. RAG: OCR ingestion, section-aware chunking, hybrid TF-IDF + semantic reranking, cited answers. Tenant self-service agent safety boundary explicitly tested -- never auto-resolves regulated decisions. LLM-as-judge CI runs on every PR.

Python · LangGraph (8-agent) · FastAPI · MCP · Hybrid RAG · OCR · SQLite · CI/CD · LLM-as-Judge

AgentProbe -- Open-Source LLM Evaluation Toolkit

github.com/venkatalakshmiparimala/AgentProbe

15-endpoint FastAPI backend routing prompts across 5 LLM providers (Anthropic, OpenAI, Azure OpenAI, Groq, Ollama) with dynamic LLM Router. Scores outputs on factual accuracy, hallucination rate, retrieval quality, completeness, and compliance. CI regression tests with golden datasets. Runs 100% locally at zero cost -- built with Claude Code daily.

Python · FastAPI · LangGraph · ChromaDB · MCP · GitHub Actions · Anthropic · OpenAI · Groq · Ollama

AI on FHIR -- Clinical NLP & LLM API Integration

github.com/venkatalakshmiparimala/AI_on_FHIR

LLM + BERT + embeddings turn English/Spanish questions into FHIR API calls -- cut manual clinical lookup time by ~60%. Role-based access control for HIPAA-style data isolation; containerized with Docker. Built trust boundary: clinical data is retrieved and displayed, never modified by AI.

Python · Flask · React · TypeScript · Docker · BERT · spaCy · LLMs · FHIR REST APIs · OAuth

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026

GPA 3.87 · NLP pipelines, anomaly detection, regression modeling (R^2=0.977), HuggingFace fine-tuning

B.Tech Computer Science & Engineering — JNTU Anantapur — 2017

TECHNICAL SKILLS

AI & LLMs: MCP (Model Context Protocol), MCP Server Development, LLM APIs (Anthropic Claude, OpenAI, Azure OpenAI, Groq, Ollama), RAG Pipelines, Hybrid Search, Embeddings, Prompt Engineering, AI Agents, LangGraph, LangChain, Human-in-the-Loop, LLM-as-Judge Evaluation, Fine-tuning (HuggingFace, PyTorch)

Python & ML: Python (9+ yrs), FastAPI, Flask, Django, Pandas, NumPy, scikit-learn, XGBoost, PyTorch, PySpark

Other Languages & Web: TypeScript, JavaScript, Node.js, React, Redux, Java, Spring Boot, PHP, GraphQL, SQL, PL/SQL, Bash

Databases & Data: PostgreSQL, MySQL, Oracle, Redis, MongoDB, SQLite, Snowflake, ETL Pipelines, Query Optimization

Cloud & DevOps: AWS (S3, EC2, RDS, Lambda, Glue, SQS, API Gateway, IAM), Azure (OpenAI, Data Factory, AKS), GCP (Vertex AI), Docker, Kubernetes, Terraform, GitHub Actions CI/CD

Security & Compliance: OAuth, HIPAA/PHI Compliance, Libsodium Encryption, AWS IAM, RBAC, Audit Logging

Practices: REST APIs, Microservices, Agile/Scrum, TDD, Code Review, Observability, Trust Boundary Design